

OUR SOLAR SYSTEM SCALE MODEL

Using Fruits and Spices to Learn About Planets!

Grade 4 Science Lesson

01. WHAT IS THE SOLAR SYSTEM?

The Solar System is our cosmic neighborhood! It consists of one big star called the **Sun**, and **eight planets** that orbit around it. Each planet is unique in size and distance from the Sun.

Some planets are tiny, like **Mercury**! Others are gigantic, like **Jupiter**! Because planets are separated by vast distances in space, we cannot see all of them with our naked eyes from Earth. Today, we will use household fruits and spices to understand these scale differences!

02. PLANET SIZES COMPARED (SCALE MODEL)

Imagine representing each planet using items from an Indian kitchen:

Category	Planet	Kitchen Model Representation
Inner Planets (Small & Rocky)	Mercury	Cardamom Pod (Elaichi) — Smallest Planet
	Venus	Gooseberry (Amla)
	Earth	Guava
	Mars	Tomato
Outer Planets (Gas Giants)	Jupiter	Watermelon — Largest Planet!
	Saturn	Large Pumpkin
	Uranus	Bottle Gourd (Lauki)
	Neptune	Coconut

03. DISTANCES BETWEEN PLANETS

- The planets in our Solar System are incredibly far apart!
- **Sun to Earth Comparison:** If Earth were the size of a small red chili pepper, the Sun (represented as a giant ball) would be located over **300 meters away**!
- Even though Earth is the third planet from the Sun, it is still millions of kilometers away in space.



Fun Solar System Fact!

Did you know? **Jupiter is so big that over 1,300 Earths could fit inside it!** Imagine filling a giant watermelon with tiny marbles — that is how big Jupiter is compared to Earth. Furthermore, Jupiter spins so fast that a single day on Jupiter lasts only about **10 hours!**

Practice Worksheet

Test your knowledge on Our Solar System Scale Model!

Student Name: _____

Date: _____

Grade: 4

1. How many planets orbit around the Sun in our Solar System?

A) 5 planets

B) 8 planets

C) 10 planets
D) 12 planets

2. Which is the smallest planet in our Solar System?

A) Earth

B) Jupiter

C) Mercury
D) Mars

3. In our scale model, which fruit represents the planet Jupiter?

A) Tomato

B) Guava

C) Gooseberry
D) Watermelon

4. Which kitchen item is used to represent our planet Earth in the lesson model?

A) Cardamom pod

B) Guava

C) Pumpkin
D) Coconut

5. How many Earths can fit inside giant Jupiter?

A) About 100

B) About 500

C) Over 1,300
D) Over 1,000

6. True or False: Inner planets are larger than outer planets.

A) True

B) False

7. How long is one day on Jupiter?

A) 24 hours

B) 10 hours

C) 365 days
D) 5 hours

8. Which vegetable is used to represent planet Saturn in the scale model?

Answer: _____

9. What object represents planet Mars in our lesson?

Answer: _____

10. Name the star at the center of our Solar System around which all planets orbit.

Answer: _____

11. Why can't we see all the planets easily with our eyes from Earth?

Answer: _____

12. List two outer planets mentioned in the scale model and the fruits/items used for them.

Answer: _____